10BONDS

A Quantitative Framework for Credit-Risk-Constrained Pareto-Efficient Bond Portfolio Construction

Multi-source term structures, heterogeneous recovery,
dependence-sensitive expected shortfall and mixed-integer optimisation

METHODOLOGICAL PRINCIPLE

The admissible portfolio set is defined by observable bond characteristics and explicitly stated risk constraints; efficient portfolios are identified through constrained multiobjective optimisation, not by ex ante rating heuristics or ad hoc weighted rankings.

Document class	Scientific methodology and model specification
Effective design	Target quantitative framework, Version 2.0
Publication date	24 October 2026
Scope	EUR and USD rated corporate and sovereign debt; investment horizons 1–30 years
Website	10bonds.com

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Abstract

We formalise the construction of cardinality-constrained, issuer-diversified fixed-income portfolios as a mixed-integer, multiobjective stochastic optimisation problem over a rating-conditioned security universe. The framework couples history-calibrated physical-measure default term structures to bond-specific loss-given-default, explicit cash-flow conventions, maturity-dependent relative-spread hurdles, latent systematic credit factors, rating-state migration and coherent portfolio tail-risk functionals. The resulting optimisation distinguishes contractual yield objectives from probabilistic return forecasts, and distinguishes marginal credit loss from dependence-sensitive portfolio loss. For each admissible yield threshold, the engine minimises discounted expected credit shortfall subject to issuer cardinality, term-matching, credit-quality and conditional expected-shortfall constraints. After exact feasibility and nondominance filtering, up to thirty economically distinct portfolios parameterise a one-dimensional user interface. Model risk is addressed through point-in-time evaluation, scenario aggregation, sensitivity analysis, numerical optimality diagnostics and reproducible calibration governance.

The theoretical construction is based on the classical notion of Pareto dominance over *the declared objectives and feasible set*; it does not imply that a model frontier is efficient with respect to unmodelled economic risks, or that historical default regularities constitute a guarantee of future results.

FORMAL OBJECTIVE

For a risk-budgeted feasible portfolio family $\mathscr{X}(H, R, C)$, identify the nondominated set

$$\mathscr{P}(H, R, C) = \mathrm{ND}\left\{\Big(\mathrm{EL}(x), -Y(x)\Big) : x \in \mathscr{X}(H, R, C)\right\}, \tag{1}$$

where H denotes the term target, R the permitted maximum rating-risk class and C the portfolio expected-shortfall budget.

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1. Mathematical architecture and notation

The platform separates *instrument eligibility*, *statistical credit estimation*, *joint portfolio risk measurement*, *discrete portfolio optimisation*, and *nondominated solution presentation*. This separation prevents market-pricing heuristics from masquerading as risk probabilities and prevents a simulation-based risk visualisation from being confused with a proof of optimiser optimality.

Let $\mathscr{I}$ be the set of eligible instruments indexed by i, and $\mathscr{G}$ the set of consolidated issuer/guarantor risk groups. Binary variable $x_i = 1$ identifies inclusion in the ten-bond portfolio, with equal nominal portfolio allocations after settlement feasibility checks. The notation used throughout is summarised below.

Symbol	Meaning	**Symbol**	Meaning
H	Chosen investment horizon or duration target (years)	R	Maximum admissible rating-risk class
Y_i	Comparable annualised security yield	D_i	Modified/effective duration, as applicable
$F_i(t)$	Physical cumulative default distribution	$\lambda_i(t)$	Physical default intensity
$\ell_i(\omega)$	Instrument loss under scenario ω	EL_i	Discounted expected credit shortfall
Z	Vector of systematic latent factors	S	Number of stochastic scenarios
CVaR_α	Conditional value-at-risk at level α	$\mathscr{X}$	Constraint-admissible binary portfolios
C	Allowable portfolio tail-loss threshold	T	Yield floor defining a frontier subproblem

Table 1: Selected notation. All loss functionals must specify horizon, exposure convention and discounting basis.

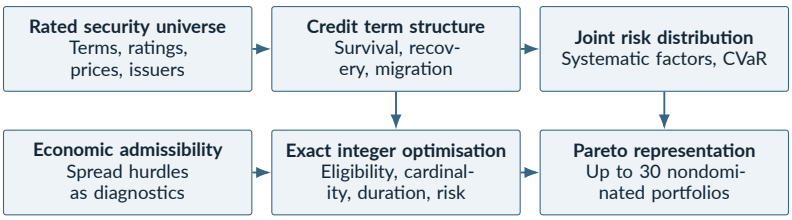


Figure 1: Target quantitative processing architecture. Arrows indicate computational inputs, not causal proof of predictive validity.

2. Instrument universe, ratings and eligibility

2.1 Universe integrity and admissible securities

The considered universe consists of EUR- and USD-denominated corporate and sovereign fixed-income instruments with at least one recognised long-term rating from one of the major international credit rating agencies. Unrated obligations are excluded; securities carrying a currently defaulted status are not admitted as performing instruments. A conservative credit-quality ceiling is obtained by mapping each agency's rating to a common ordinal risk class and taking the worst observed class:

$$R_i = \max_{a \in \mathscr{A}_i} \rho\left(r_i^{(a)}\right), \quad \rho : \{\text{agency grades}\} \longrightarrow \{1, \dots, 10\}. \tag{2}$$

The user-facing classes span AAA (class 1), AA (2), A (3), BBB+ (4), BBB (5), BBB– (6), BB (7), B (8), CCC (9) and CC/C (10); defaulted D obligations are separately excluded. Rating notches within the broader AA, A, BB and B families are preserved where available in the credit term-structure layer. A deterministic mapping table is version-controlled, including rules for split ratings, missing ratings and issuer-versus-issue rating applicability.

Additional exclusions comprise minimum trading denominations above 200,000 units, unresolvable issuer identity, economically inconsistent cash-flow schedules, non-executable pricing, and statistically flagged yield anomalies after quote-quality adjudication. Statistical deviation *alone* is insufficient evidence that a high-yielding security is erroneous: stale prices, distressed market pricing and option effects are classified separately.

2.2 Term coherence and execution feasibility

The individual instrument duration screen is

$$0.8H \leqslant D_i \leqslant 1.2H, \tag{3}$$

while portfolio duration obeys

$$\left|D_p - H\right| \leqslant 0.05H, \qquad D_p = \frac{1}{10}\sum_i D_i x_i. \tag{4}$$

Modified duration is employed for option-free fixed-rate debt; effective duration is required where cash flows depend on rate-sensitive embedded options. For a maturity-defined consumer horizon, a separate cash-flow compatibility assessment is necessary because Macaulay/modified duration does not equal the contractual redemption date. Actual denomination rules and settlement amounts are checked against the nominal equal-allocation convention.

2.3 Currency-consistent comparability

EUR and USD bonds are not compared through raw nominal yield differences. A comparison is performed either within the investor's selected currency or on a currency-hedged, conventionally matched basis. With a hedged return approximation,

$$Y_i^{\text{home,hedged}} \approx Y_i^{\text{local}} + c_i^{\text{FX}} - c_i^{\text{transaction}}, \tag{5}$$

where hedge carry and costs are tenor- and instrument-specific. This is a first-order decomposition, not a substitute for complete hedge cash-flow valuation. Currency-conditional benchmarks are maintained separately throughout spread estimation.

3. Multi-source physical credit-risk term structures

3.1 Survival representation and estimation

Let τ_i denote default time, with physical survival $S_i(t)$ and cumulative default probability $F_i(t)$:

$$F_i(t) = \mathbb{P}^{\mathbb{P}}(\tau_i \leqslant t) = 1 - S_i(t), \tag{6}$$

$$S_i(t) = \exp\left[-\int_0^t \lambda_i(u)\,du\right], \qquad \lambda_i(t) \geqslant 0. \tag{7}$$

Default probabilities are our own estimation based on historical data, all harmonised by issuer definition, rating cohort, default-event taxonomy, seasoning, geography, currency basis, sample horizon, withdrawal/censoring treatment and observation window. Corporate and sovereign exposures are estimated with separate source populations rather than transferring corporate default rates directly to sovereign issuers [1, 2].

For source a, rating r and cohort horizon t, let $\widehat{F}_{a,r}(t)$ denote the empirical cumulative default estimate. A reproducible pooled estimator may take the form

$$\widehat{F}_r(t) = \sum_{a \in \mathscr{A}_r} w_{a,r}\widehat{F}_{a,r}(t), \quad w_{a,r} \geqslant 0, \quad \sum_a w_{a,r} = 1, \tag{8}$$

with documented weighting and fixed weights across time for a given cohort definition. This convex mixture preserves nondecreasing cumulative distributions when all component estimates are monotone. Separate monotonicity and ordinal regularisation may be applied where sampling noise creates counterintuitive grade crossovers:

$$\widehat{F}_r(t) \leqslant \widehat{F}_{r+1}(t), \qquad \frac{\partial \widehat{F}_r(t)}{\partial t} \geqslant 0. \tag{9}$$

These restrictions are statistical calibration choices; empirical crossovers caused by small samples must be documented rather than silently erased. Interpolation is performed in cumulative hazard space $\Lambda_r(t) = -\log[1 - F_r(t)]$, preserving nonnegative integrated hazard. Beyond the observed sample horizon, extrapolation uses regularised tails, uncertainty envelopes and explicit 11–30-year validation flags.

3.2 Rating migration and absorbing-state dynamics

Rating transitions are modelled separately from unconditional default frequencies. A finite-state Markov approximation employs one-year row-stochastic transition matrix P with default as an absorbing state; in a continuous-time representation, a generator Q satisfies $Q\mathbf{1} = 0$ and

$$P(t) = \exp(tQ), \qquad P_{DD}(t) = 1. \tag{10}$$

Multi-period migration distributions condition downgrade-sensitive bond valuations and refinancing scenarios. Empirical transition matrices and cumulative default cohorts cannot be freely interchanged; calibration must reconcile the two representations or explicitly state their distinct purposes.

3.3 Physical versus risk-neutral measures

The statistical distribution $\mathbb{P}$ used for expected loss must not be equated to the pricing measure $\mathbb{Q}$. A market spread incorporates expected credit loss, covariance with marginal utility, bond liquidity, tax, optionality and transaction effects. In a reduced-form approximation,

$$s_i^{\text{market}}(t) \simeq \underbrace{\text{ELrate}_i^{\mathbb{P}}(t)}_{\text{physical loss}} + \underbrace{\pi_i^{\text{credit}}(t)}_{\text{risk premium}} + \pi_i^{\text{liq}}(t) + \pi_i^{\text{opt}}(t) + \varepsilon_i(t). \tag{11}$$

The expression is a risk-premium decomposition, not an exact no-arbitrage pricing identity. No risk-neutral implied default likelihood is presented as an empirically observed frequency.

4. Bond-level exposure, recovery and credit shortfall

4.1 Contract-contingent loss severity

Default loss is estimated at security rather than merely issuer level. Recovery is differentiated by claim seniority, enforceable security, structural subordination, governing law, guarantees and sovereign restructuring characteristics. Let $EAD_i(t)$ denote remaining economically at-risk exposure and $R_i(t, \omega)$ its recovery fraction. Discounted credit shortfall is

$$\ell_i(\omega) = \frac{1}{P_i^0} \int_0^{H_i} B_{c_i}(0, t)\, EAD_i(t, \omega)[1 - R_i(t, \omega)]\, dN_i(t, \omega), \tag{12}$$

where $N_i(t) = \mathbf{1}_{\{\tau_i \leqslant t\}}$ is a single-jump default indicator, B_{c_i} is the matching-currency discount factor and P_i^0 the current invested present value. The integral applies at default for a single-jump process. Missed coupons, accelerated recovery, post-default delays and accrued interest are incorporated in the exposure convention when material.

Then

$$\text{EL}_i = \mathbb{E}^{\mathbb{P}}[\ell_i], \qquad \text{EL}_p(x) = \frac{1}{10} \sum_i x_i \text{EL}_i. \tag{13}$$

For fixed exposure, deterministic recovery and a common terminal loss date, this reduces to $\text{EL}_i \approx \text{PD}_i(H_i)\text{LGD}_i$, with $\text{LGD}_i = 1 - R_i$. That familiar expression is therefore a *special case*, not the generic

pricing model. The formerly illustrative 60% loss severity is superseded by security-conditional recovery calibration and sensitivity bounds [3].

4.2 Yield conventions and cash-flow uncertainty

The instrument return target Y_i is a comparable annualised contractual yield measure, generally yield-to-maturity for option-free bonds and appropriately computed yield-to-worst with option-aware scenarios for redeemable bonds. For price P_i and deterministic cash flows CF_{ik},

$$P_i = \sum_{k=1}^{n_i} \frac{CF_{ik}}{(1 + Y_i/m_i)^{m_i t_{ik}}}, \tag{14}$$

using its specific compounding/day-count convention. For callable issues, alternative redemption branches are explicitly enumerated; yield-to-worst is a conservative quotation convention but not an expectation under a stochastic exercise model. The average displayed yield,

$$Y_p(x) = \frac{1}{10} \sum_i Y_i x_i, \tag{15}$$

remains distinct from the stochastic realised return $R_p(\omega)$ and from $\mathbb{E}[R_p]$. A quoted 4% yield is not a promise of 4% realised return.

5. Maturity-dependent relative credit-spread discipline

5.1 Economic hurdle system

For a rating transition from stronger class $r-1$ to weaker class r, define the maturity-dependent incremental yield benchmark $\delta_r(t)$ in basis points. The term matrix is an *economic adequacy diagnostic*: a lower-rated instrument is marked as adequately compensated only where the observed comparable spread differential meets the relevant hurdle. The matrix is never used to remove a portfolio solely because a weaker-rated bond's premium fails a heuristic threshold; such removal could invalidate global Pareto efficiency.

Rating transition	1Y	3Y	5Y	7Y	10Y
AAA → AA	5	8	10	12	15
AA → A	10	12	15	20	25
A → BBB+	20	28	35	47	60
BBB+ → BBB	12	16	20	29	40
BBB → BBB−	25	37	50	70	95
BBB− → BB	40	55	70	100	140

Table 2: Published target rating-by-maturity minimum incremental-spread configuration (basis points). These parameter values are design-policy hurdles; they are not historical default estimates or statistically guaranteed optimal spreads.

For integer/non-integer maturities between knots t_k and t_{k+1}, the piecewise-linear specification is

$$\delta_r(t) = \delta_r(t_k) + \frac{t - t_k}{t_{k+1} - t_k}[\delta_r(t_{k+1}) - \delta_r(t_k)]. \tag{16}$$

The 11–30-year portion and grades below BB are separately estimated using horizon-specific hazard, recovery, migration and term-premium evidence, rather than indiscriminate linear extrapolation. A term structure need not be increasing at every rating; speculative-grade annualised loss compensation can be hump-shaped or inverted.

5.2 From incremental actuarial loss to spread premia

A rough first-order expected-loss differential over t years is

$$\delta^{\mathrm{EL}}_{r,t} \simeq 10^4 \, \frac{\mathrm{LGD}_r F_r(t) - \mathrm{LGD}_{r-1} F_{r-1}(t)}{t} \quad \text{bp per annum.} \tag{17}$$

A more faithful default-risk component solves for the spread differential that equalises risky present values against the appropriate benchmark annuity. The policy hurdle is then decomposed as

$$\delta_r(t) = \delta_r^{\mathrm{EL}}(t) + \delta_r^{\mathrm{systematic}}(t) + \delta_r^{\mathrm{liquidity}}(t) + \delta_r^{\mathrm{structure}}(t). \tag{18}$$

The components may be estimated jointly; blindly summing empirically estimated premia from non-orthogonal datasets risks double counting. Benchmark curves and calibration sets are currency-specific.

At five years, the defined absolute spread reference over the appropriate EUR benchmark reproduces AAA 35, AA 45, A 60, BBB+ 95, BBB 115, BBB– 165 and BB 235 bp. At other horizons, let

$$s_r^{\mathrm{floor}}(t,c) = s_{AAA}^{\mathrm{floor}}(t,c) + \sum_{k=2}^{r} \delta_k(t,c). \tag{19}$$

The relative matrix is *not* added again to the established absolute floors; the latter are recursive sums of the former. A EUR Bund-derived anchor must not be automatically imposed on USD issues.

5.3 A minimal pairwise illustration

Consider an AA bond at 4.00%, and otherwise comparable A bonds at 4.05% and 4.20%. The AA-to-A hurdle is 12, 15 and 25 bp at 3, 5 and 10 years, respectively. The 4.05% instrument falls below all three thresholds, whereas the 4.20% instrument passes at three and five years but falls short at ten. These labels signal relative compensation, *not binding exclusion*: subject to other portfolio constraints, the 4.20% security remains admissible at ten years, and may be essential to an efficient portfolio meeting a specified yield floor.

6. Dependence-sensitive stochastic portfolio losses

6.1 Systematic factor construction

Independently drawn issuer default times can materially understate joint downside in recessions. The target risk engine employs latent common factors for global credit conditions, issuer domicile, sector, sovereign funding, commodity exposures and relevant macro-financial regimes. A reduced-form factor representation is

$$\lambda_i(t \mid Z_t) = \lambda_i^0(t) \exp\left(\beta_i^{\mathsf{T}} Z_t - \frac{1}{2}\beta_i^{\mathsf{T}} \Sigma_Z \beta_i\right), \tag{20}$$

where the normalising term is a useful lognormal-scale adjustment, not sufficient by itself to guarantee that unconditional cumulative default curves match all marginal calibration targets. Marginal survival reconciliation is a separate numerical calibration step.

Conditional on macro factors and issuer-level random effects, event-time simulation permits correlated default indicators

$$I_i(H) = \mathbf{1}_{\{\tau_i \leqslant H\}}, \qquad \mathbb{P}(I_i = 1, I_j = 1) \neq \mathbb{P}(I_i = 1)\mathbb{P}(I_j = 1). \tag{21}$$

A Gaussian or Student-t copula is an alternative construction when joint dependence is calibrated through latent asset values. Heavy-tail dependence is sensitivity-tested instead of assumed universally. Recoveries are allowed to deteriorate in common stress states, rather than remaining mechanically independent of default clustering [7, 4].

6.2 Scenario portfolio loss and coherent expected shortfall

For scenario $s \in \{1, \dots, S\}$, portfolio loss is

$$L_s(x) = \frac{1}{10} \sum_i x_i \ell_{is}. \tag{22}$$

The tail risk measure at confidence $\alpha = 0.95$ is

$$\mathrm{CVaR}_\alpha(L) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1-\alpha} \mathbb{E}[(L-\eta)_+] \right\}, \tag{23}$$

which is coherent under standard conditions and remains well-defined for distributions with probability atoms [5]. In the equal-weight scenario approximation,

$$\widehat{\mathrm{CVaR}}_\alpha(x) = \min_{\eta\in\mathbb{R}} \left\{ \eta + \frac{1}{(1-\alpha)S} \sum_{s=1}^{S} \max(L_s(x) - \eta, 0) \right\}. \tag{24}$$

The stochastic layer also computes any-default frequency, the distribution of default counts, probability of principal shortfall, discounted loss quantiles and losses conditional on systematic shocks. These are distinct indicators: a *default event* does not necessarily imply a negative total holding-period return, and a negative sale-period return can occur without any default.

6.3 Simulation protocol and numerical error

The reporting distribution uses a target of one million Monte Carlo scenarios per specified portfolio risk horizon. Scenario sets used within mixed-integer optimisation can be reduced, stratified or otherwise compressed for tractability; shortlisted solutions are then re-evaluated against the larger independent validation sample. A sample of 10^6 paths reduces sampling noise but does not correct erroneous marginal PDs, wrong default correlations or misspecified recoveries. If $\widehat{p}$ is a simulated event frequency, its approximate independent-sampling standard error is $\sqrt{\widehat{p}(1-\widehat{p})/S}$; weighted or dependent scenario schemes require corresponding variance estimators.

7. Exact cardinality-constrained portfolio optimisation

7.1 Admissible binary set

Each portfolio contains precisely ten bonds, at most one obligation per consolidated issuer risk group, and at most two issues in the B+ or lower part of the rating scale. Let $g(i)$ denote the issuer group and $\mathscr{B}$ the securities rated B+ or weaker. A central feasible set is

$$\begin{aligned}
\mathscr{X}(H,R,C) = \Big\{ x \in \{0,1\}^{|\mathscr{I}|} :\ & \sum_i x_i = 10, && (25)\\
& \sum_{i:g(i)=g} x_i \leqslant 1 \quad \forall g \in \mathscr{G}, && (26)\\
& \sum_{i\in\mathscr{B}} x_i \leqslant 2, && (27)\\
& x_i = 0 \quad \text{if } R_i > R, && (28)\\
& |D_p(x) - H| \leqslant 0.05H, && (29)\\
& \mathrm{CVaR}_{.95}(L(x)) \leqslant C, && (30)\\
& A_k(x) \leqslant b_k \quad \forall k \in \mathscr{K} \Big\}. && (31)
\end{aligned}$$

Here A_k encompasses explicit concentration, sector, domicile, sovereign/corporate mix, liquidity or structural limits when activated by policy; each active constraint and threshold is stored in the run manifest. Individual duration bounds and eligibility filters are applied upstream. Feasibility takes priority over forcing a ten-bond result: if no feasible set exists, the platform must report infeasibility and explain the blocking conditions.

7.2 Two-objective constrained frontier generation

For each contractual-yield threshold T define

$$\boxed{x^\star(T) \in \arg\min_{x\in\mathscr{X}(H,R,C)} \mathrm{EL}_p(x) \quad \text{subject to} \quad Y_p(x) \geqslant T.} \tag{32}$$

Among minimisers of identical expected loss, choose the highest Y_p as a lexicographic secondary criterion. This ε-constraint formulation avoids opaque linear combinations of yield and expected loss. It is exact *relative to the finite eligible universe, defined scenario approximation and imposed constraints*, up to a disclosed mixed-integer optimality tolerance.

Proposition 1 (Conditional weak Pareto optimality)**.** *If $x^\star(T)$ is globally optimal for* (32)*, no feasible portfolio with yield at least $Y_p(x^\star)$ can have* strictly lower *expected loss. Lexicographically breaking loss ties toward higher yield excludes weakly dominated equal-loss solutions.*

Proof. Were a feasible y to have $Y_p(y) \geqslant Y_p(x^\star)$ and $\mathrm{EL}_p(y) < \mathrm{EL}_p(x^\star)$, then $Y_p(y) \geqslant T$ and y would improve the objective in (32), contradicting global optimality. Equal-loss higher-yield alternatives are eliminated by the stated tie-breaking rule. □

The proposition *does not* claim efficiency against omitted market risk, and a frontier constrained by CVaR $\leqslant C$ is a two-objective frontier *conditional* on that tail budget. A simultaneous unconstrained ordering over (EL, $-Y$, CVaR) is three-dimensional and has a distinct nondominated set.

7.3 Linearisation of expected shortfall constraints

In a reduced scenario set of S_o paths, impose auxiliary variables $\xi_s \geqslant 0$ and $\eta \in \mathbb{R}$:

$$\eta + \frac{1}{(1-\alpha)S_o} \sum_{s=1}^{S_o} \xi_s \leqslant C, \tag{33}$$

$$\xi_s \geqslant L_s(x) - \eta, \qquad \xi_s \geqslant 0. \tag{34}$$

This represents the empirical CVaR budget through linear constraints. Combined with binary selection, issuer constraints and duration bounds, it yields a mixed-integer linear programme when scenario losses are precomputed and additive by security. Correlated risk enters through the *joint scenario generation*, not through an incorrect additive approximation to portfolio CVaR. Nonlinear refinancing, position-dependent market impact or endogenous hedging necessitate richer formulations.

7.4 Why the spread hurdle cannot be a global hard filter

A hard hurdle $Y_j - Y_i \geqslant \delta_{r,t}$ is not a dominance test when i and j differ in default risk, maturity, issuer or portfolio feasibility. Deleting the lower-rated issue below a pricing hurdle can eliminate a portfolio which is efficient under (1). Therefore the spread matrix is used for valuation diagnostics, quality labelling, audit signals and sensitivity analysis; it does not amend $\mathscr{X}$ solely on heuristic pricing grounds. Similarly, a bond can be discarded by pairwise dominance only where substitution preserves every active feasibility constraint and weakly improves every modelled objective.

8. The 30-position efficient-set user representation

The user interface exposes a single slider with ten major credit-risk classes and three within-class return/risk gradations (low, intermediate, high), targeting thirty positions. Underlying construction uses the risk ceiling R, a family of yield thresholds $T_{R,j}$ and the risk budget C. Eligible candidate portfolios are solved, independently re-evaluated and subjected to nondominance filtering. Portfolio solutions are sorted by expected credit loss (with secondary return ordering), while the major class identifies the *maximum permitted* bond grade, not a requirement to own the riskiest grade.

Let $\mathscr{S}$ be the candidate union and define

$$\mathrm{ND}(\mathscr{S}) = \left\{ x \in \mathscr{S} : \nexists y \in \mathscr{S} : \ \mathrm{EL}(y) \leqslant \mathrm{EL}(x),\ Y(y) \geqslant Y(x),\ \text{one strict} \right\}. \tag{35}$$

Duplicates and dominated candidates are suppressed. When a class contains fewer than three distinct admissible efficient portfolios, a displayed position cannot be fabricated merely to preserve an exact count. The interface may dynamically expose fewer than thirty distinct solutions or transparently map multiple stops to the same economically identical portfolio. The front-end's simplicity therefore does not relax the mathematical definition of efficiency.

9. Second-order risk: migration, liquidity and market repricing

Under a hold-to-contractual-redemption assumption, coupon and principal impairment dominates permanent credit loss. Under a fixed liquidation date, interest-rate and spread repricing are also

relevant. For an option-free bond, the standard local approximation is

$$\frac{\Delta P_i}{P_i} \approx -D_i^{\text{rates}} \Delta r - D_i^{\text{spread}} \Delta s_i + \frac{1}{2}\Gamma_i (\Delta y)^2, \tag{36}$$

with appropriate higher-order and cross terms omitted from the local expansion. Rating migration affects both default intensities and repricing/spread risk, but the cash redemption itself is not automatically reduced by a downgrade absent an actual impairment or contractual trigger.

Additional state variables include subordinated-claim severity, executable bid–ask spreads, market depth, callable/putable exercise behaviour, extension risk, and issuer domicile/sector concentration. These variables are not collapsed into a scientifically unsubstantiated universal weighted score. Instead they enter instrument eligibility, scenario losses, hard diversification limits, uncertainty analysis or clearly defined auxiliary risk budgets [4].

10. Statistical verification, numerical assurance and backtesting

10.1 Calibration diagnostics

Rating-term PD estimates are checked for (i) survival monotonicity; (ii) grade-ordering consistency; (iii) coherent treatment of censoring, withdrawals and repeated defaults; (iv) calibration stability across historical windows; and (v) rating migration versus cohort default consistency. Proper scoring rules, reliability plots, out-of-time Brier scores and discrimination statistics may be reported for default-event forecasting where sufficient issuer-level observations exist. Calibration error must not be disguised by averaging sparse samples.

10.2 Point-in-time historical reconstruction

Backtests form a sequence of information sets $\mathscr{F}_{t_0}$ using only securities, ratings, quotes, legal terms and issuer groups available at decision date t_0. Return paths follow actual coupons, recovery events, early redemptions, market liquidity and any rebalancing rule. Survivorship, look-ahead, stale-quote and inception biases are specifically controlled. Candidate portfolios are benchmarked against duration-matched, currency-matched and rating-constrained portfolios, plus appropriate investable fixed-income indices where constituents are available.

10.3 Numerical optimiser certification

On tractable sub-universes, exhaustive enumeration provides a reference efficient set against which the MILP output is compared. Larger runs record incumbent objectives, best bounds, integrality gaps, infeasibility certificates, rounding tolerances and scenario-sampling error. A computed solution is described as *globally optimal for the model* only when the solver provides the corresponding certificate within its declared tolerance; the phrase does not imply globally best realised performance in financial markets.

10.4 Regime stress and sensitivity diagnostics

Stress dimensions comprise recession, financial crisis, sovereign restructuring, regional conflict, energy-price shocks, supply-chain disruption, sudden risk-free rate repricing, downgrade cascades, funding closure and liquidity evaporation. One-at-a-time and joint parameter perturbations are evaluated for PD, default-factor correlations, conditional LGD, bid–ask costs, yield-floor assumptions, FX hedge carry and CVaR budgets. Robustness can be expressed through a stability functional

$$\mathscr{R}(x) = 1 - \frac{1}{|\Theta|} \sum_{\theta \in \Theta} \frac{\left|\text{supp}(x) \square \text{supp}(x_\theta^\star)\right|}{20}, \tag{37}$$

where the symmetric-difference distance measures portfolio membership changes under alternative plausible calibrations. This is a *portfolio stability diagnostic*, not an additional verified prediction statistic.

11. Governance, version control and reproducibility

Every ranking run is tied to a deterministic methodology version and a data snapshot. The provenance manifest specifies: quote timestamp and venue convention; rating source and mapping revision; historical default-study vintages; hazard interpolation and extrapolation scheme; LGD/recovery assumptions; currency curve; spread-hurdle matrix revision; Monte Carlo random seed and factor parameters; optimisation solver version and tolerances; scenario compression; and active exposure limits. Analytic reproducibility requires re-running the same portfolio construction against the same parameter snapshot, not against a subsequently refreshed bond database.

A publication change log distinguishes (a) policy-parameter updates, (b) statistical recalibration, (c) security universe/quote coverage improvements and (d) modifications of the mathematical optimiser. Independent model review should verify the implemented algorithms against the published specification before any claim of external assurance is made. Methodology transparency does not require disclosure of proprietary issuer-level datasets or source material beyond permitted licences.

12. Interpretation boundaries and methodological limitations

1. Pareto efficiency is *conditional*: it concerns the objectives, finite eligible universe, estimation conventions and explicit constraints. It is not a proof of perfect security pricing or future superiority.
2. Probability models depend on historical credit regimes and forward-looking calibration assumptions. Rare systemic crises, long-horizon sovereign restructuring, regulatory changes and correlated recoveries may be insufficiently represented.
3. Yield-to-worst, expected loss and tail probability describe different random or contractual objects. They must never be numerically interchanged or interpreted as guarantees of net investor return.
4. A single lender or issuer may have many economically dependent securities and affiliates; group-level concentration controls reduce but cannot eliminate systemic overlap.
5. Bond prices, ratings, spreads and market depth can change immediately. A published ranking is time-stamped and cannot establish that the same portfolio remains executable later.
6. The methodological design is a quantitative tool, not an offer, credit rating, brokerage facility, capital guarantee, suitability assessment, or personalised investment recommendation.

Conclusion

The 10bonds framework is designed as a formal discrete-optimisation architecture combining empirical physical default term structures, cash-flow-sensitive severity estimates, multi-factor dependence, coherent tail-loss constraints and auditable cardinality-constrained efficient-set computation. The rating-by-maturity relative-spread surface acts as a distinct financial-economic diagnostic rather than an expedient rejection rule. Mathematical rigor lies in preserving the boundary between measurable inputs, modelling assumptions, solver guarantees and realised financial outcomes. The resulting portfolio set offers a transparent basis for comparing the trade-off between *indicated annualised yield* and *modelled capital-loss exposure*, with all conclusions strictly conditional on a dated calibration and admissible universe.

A. Reproducible policy parameter register

Policy dimension	Specified design value / rule
Eligible asset classes	Rated corporate and sovereign fixed-income instruments in EUR and USD
Permitted investor horizon	1–30 years
Portfolio cardinality	10 bond lines
Group concentration	Maximum one security per consolidated issuer risk group
Lower-grade concentration	Maximum two lines rated B+ or below
Eligibility rating criterion	At least one long-term rating from a major rating agency; worst mapped risk class is the ceiling measure
Defaulted debt	Excluded from standard performing universe
Minimum denomination	Exclude minimum investment sizes above 200,000 currency units
Single-security duration	Within ±20% of chosen H
Portfolio duration	Within ±5% of chosen H
Primary credit objective	Minimise expected discounted credit shortfall at prescribed yield floor
Tail constraint	CVaR at 95% confidence subject to explicit risk budget C
Simulation	Up to 10^6 reporting/validation scenarios; smaller calibrated scenario sets permissible during optimisation
Rating/spread discipline	Table 2 for 1, 3, 5, 7, 10 years; intermediate-maturity interpolation; separate long-term and high-yield calibration
Portfolio interface	One slider; 10 major credit-risk ceilings × 3 internal gradations; up to 30 distinct admissible frontier points
Fair comparison	Currency-consistent yield/hedged yield, comparable instrument cash-flow and option conventions
Output certification	Feasibility, global nondominance filter, declared MILP optimality tolerance and scenario uncertainty

Table 3: Methodology configuration defined in this paper. Empirical parameters and limits not stated numerically here must be taken from versioned configuration, not silently presumed universal.

B. Proof obligations and numerical control checklist

1. **Probability validity:** for every modelled rating and security, $F(0) = 0$, $0 \leqslant F(t) \leqslant 1$, and $F(t_1) \leqslant F(t_2)$ for $t_1 \leqslant t_2$.
2. **Issuer independence of names:** the ten allocations are to economically distinct consolidated risk groups, not merely different ISINs or legal names.
3. **Cash-flow integrity:** instrument value reproduces the quoted clean/dirty price within a defined tolerance using documented accrual conventions.
4. **Risk coherence:** $\mathrm{CVaR}_\alpha(L) \geqslant \mathbb{E}[L]$ for the same loss distribution and standard α where applicable; empirical CVaR must not be added bondwise.
5. **Constraint integrity:** every returned portfolio satisfies its recorded rating ceiling, issuer count, denomination, low-grade count, duration and risk budget.
6. **Frontier dominance:** for each displayed portfolio x, there is no candidate y with $\mathrm{EL}(y) \leqslant \mathrm{EL}(x)$ and $Y(y) \geqslant Y(x)$ with one strict inequality, under identical model inputs.
7. **Numerical audit:** optimisation bounds, scenario set hash, data snapshot and model seed permit independent reproduction of each reported portfolio.
8. **Market reproducibility:** quote cut-offs, stale-price exclusions and actual transactable denominations prevent theoretical zero-lot portfolios from being represented as executable portfolios.

C. Selected scientific and institutional references

References

[1] 10bonds (2026). Own estimation of corporate default and rating transition probabilities based on public historical data, harmonised by cohort and horizon. Relevant: static-pool methodology, rating transitions and multi-year cumulative issuer default rates.

[2] 10bonds (2026). Own estimation of sovereign default and rating transition probabilities based on public historical data, kept separate from corporates. Relevant: sovereign-specific ratings, default conventions and cohort methodology.

[3] Basel Committee on Banking Supervision (2026 compilation). *Basel Framework, Credit Risk — Internal Ratings-Based Approach*. Basel framework. Relevant: PD, LGD, exposure and maturity risk components, and model estimation standards.

[4] Basel Committee on Banking Supervision (2026 consolidated guidelines). *General Credit Risk Management*, CRI10. BIS publication. Relevant: concentrations, macroeconomic stress, governance and credit risk measurement.

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[6] Rockafellar, R. T., and Uryasev, S. (2000). "Optimization of Conditional Value-at-Risk." *The Journal of Risk*, 2(3), 21–41. Author's publication index.

[7] International Monetary Fund (2006). *Review and Implementation of Credit Risk Models of the Financial Sector Assessment Program*. Working Paper WP/06/134. IMF working paper.

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